TED UNIVERSITY

Department of Computer Engineering

MoonAI

Test Plan Report

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1 Introduction

This document defines the current test plan for **MoonAI**, a modular predator-prey simulation platform used to study continuous neuroevolution with NEAT. The report is written against the repository state that exists today and intentionally describes only testing assets, workflows, and verification activities that are already implemented in the codebase, build system, or project infrastructure.

MoonAI contains several verification-relevant subsystems: C++ simulation and evolution libraries, an SFML-based visualization frontend, an optional CUDA acceleration path, command-line experiment orchestration, structured output logging, a Python analysis pipeline, a profiler executable, and GitHub Actions workflows for automated build and test execution. The test plan therefore has to cover more than unit-level algorithm checks; it must also describe end-to-end execution, artifact generation, performance observation, and manual user-oriented smoke testing.

1.1 Purpose

The purpose of this test plan is to:

- identify the MoonAI items currently under test,
- describe the features that are verified by automated and manual checks,
- define the test methodologies that are already implemented,
- assign responsible personnel for each testing activity,
- record test control procedures, schedule, risks, and deliverables.

1.2 Document Baseline

The report is based on the following implemented project assets.

Table 1: Implemented verification assets referenced by this report

Asset	Use in testing
`tests/test_evolution.cpp`	Automated GoogleTest coverage for NEAT data structures, mutation, crossover, speciation, JSON round-trip, and neural network activation behavior.
`tests/test_simulation_sensors.cpp`	Automated GoogleTest coverage for simulation sensor encoding and density calculations.
`tests/CMakeLists.txt`	Registers the test executable with GoogleTest discovery so tests can run through CTest.
`just test / ctest`	Local developer entry point for automated regression execution.
`.github/workflows/ci.yml`	Cross-platform automated build and test execution on Linux and Windows in Debug and Release configurations.
`.github/workflows/release.yml`	Release build verification including automated test execution before packaging artifacts.
`src/main.cpp and src/app.cpp`	Runtime path for configuration validation, CLI orchestration, headless or GUI execution, CPU or GPU selection, and output generation.
`analysis/moonai_analysis/`	Post-run analysis pipeline that converts experiment outputs into a self-contained HTML report.
`profiler/moonai_profiler/ and src/profiler_main.cpp`	Built-in performance test and timing-report path for frame-time and scoped performance inspection.
`web/index.html and .github/workflows/pages.yml`	Public publication path for generated PDF reports and project documents.

1.3 Goals

The implemented testing approach focuses on the following goals:

- correctness of the highest-risk algorithmic logic in evolution and sensing,
- safe handling of configuration values and command-line overrides,
- stable end-to-end execution in GUI, headless, CPU-only, and optional GPU-enabled modes,
- integrity of generated output artifacts such as `config.json`, `stats.csv`, `species.csv`, and `genomes.json`,
- repeatable build and regression execution on both Linux and Windows,
- visibility into runtime performance through the profiler and optional Nsight tooling,
- internal user-oriented acceptance checks before publishing project deliverables.

2 Test Items and Scope

The system under test is not a single executable only; it is the complete MoonAI project distribution used during development, experimentation, profiling, and report generation.

2.1 Primary Test Items

Table 2: Primary items under test

Item	Source path	Verification focus
Core configuration and CLI	src/core/	Configuration serialization, validation rules, command-line parsing, and override handling.
Simulation engine	src/simulation/	Agent sensing, state updates, headless execution, and interaction with evolution.
Evolution engine	src/evolution/	Innovation tracking, mutation, crossover, species logic, and neural network execution.
Application runtime	src/app.*	Startup, headless or GUI branching, CPU or GPU enablement, reporting cadence, and graceful shutdown.
Visualization	src/visualization/	Manual smoke verification of rendering, controls, and agent inspection flows.
GPU acceleration path	src/gpu/	Optional CUDA execution and CPU fallback behavior when GPU is disabled or unavailable.
Structured logging	src/core/logger.*	Creation and format of run directories and machine-readable output files.
Analysis pipeline	analysis/moonai_analysis/	Conversion of completed experiment outputs into a self-contained HTML report.
Profiler pipeline	profiler/moonai_profiler/	Timed run capture, aggregation, and HTML profiling report generation.
Website publication path	web/ and Pages workflow	Public access to generated course-report PDFs from the project site.

2.2 In Scope

The following verification activities are in scope because they are already supported by the repository:

- automated unit tests registered through GoogleTest and CTest,
- automated cross-platform CI build-and-test execution,
- manual and scripted end-to-end runs of the main application,
- validation of command-line interfaces and configuration checking,

- verification of generated experiment, analysis, and profiler artifacts,
- internal manual acceptance and closed beta smoke checks by the three team members,
- publication of the final report PDF through the existing website pipeline.

2.3 Out of Scope

The following items are explicitly out of scope for this report because no corresponding formal implementation exists in the current project:

- external public beta distribution or telemetry collection,
- cloud deployment testing,
- exhaustive CUDA hardware certification across multiple GPU models,
- security penetration testing,
- networked or distributed simulation testing,
- automated GUI interaction testing through a UI automation framework.

3 Features To Be Tested

This section lists the implemented product features that are covered by current testing activities. The emphasis is on features that already have executable verification paths, either through automated commands or through manual smoke procedures supported by the application.

3.1 Feature Coverage Matrix

Table 3: Implemented features currently covered by testing

Feature group	Covered behavior	Current verification mode
`Config and CLI`	Validation of numeric bounds, handling of invalid overrides, experiment listing, experiment selection, runtime naming, headless mode, and CPU-only mode.	Unit plus integration plus manual
`Evolution correctness`	Innovation numbering, mutation behavior, crossover behavior, species compatibility, network activation, and JSON round-trip behavior.	Automated unit tests
`Simulation sensing`	Nearest-entity delta encoding, missing-target sentinels, and local density values in sensor vectors.	Automated unit tests
`Application startup`	Construction of simulation and evolution managers, optional visualization setup, signal handling, and final report logging.	Integration plus manual system
`Output generation`	Creation of run directory, configuration snapshot, metrics CSV, species CSV, genome JSON, and final flush at shutdown.	Integration plus manual artifact inspection
`Cross-platform build stability`	Build and test execution on Linux and Windows in Debug and Release configurations.	GitHub Actions CI
`Analysis reporting`	Discovery of completed runs and generation of a self-contained HTML analysis report.	Integration
`Profiler reporting`	Capture of scoped timing data, generation of profiler JSON, and HTML performance report output.	Performance test workflow
`Visualization usability`	Launch, pause or resume, stepping, zoom, camera motion, agent selection, and graceful exit.	Manual acceptance
`GPU fallback path`	Runtime execution with optional GPU enablement and explicit `--no-gpu` fallback to CPU.	Manual system plus performance
`Document publication`	Inclusion of generated PDFs under `docs/` and publication through GitHub Pages.	Manual release smoke check

3.2 Features Explicitly Not Covered by Current Tests

Current tests do not attempt to prove the following because the supporting infrastructure is not present in the repository:

- usability studies with external non-team users,
- load balancing or distributed experiment execution,
- long-duration soak testing across the full 330-run matrix on every target platform,
- correctness claims for third-party dependencies beyond successful integration,

- GUI regression capture through screenshot-diff tooling,
- external email delivery of course documents.

4 Testing Methodology

MoonAI uses a layered testing approach. The implemented methodology starts with automated unit tests for algorithmic correctness, extends to integration and system-level command execution, and then adds performance observation and internal manual acceptance checks.

4.1 Unit Testing

Automated unit testing is implemented with GoogleTest and registered through `gtest_discover_tests()` in `tests/CMakeLists.txt`. The current test suite covers the highest-risk deterministic logic:

- innovation number allocation and reset behavior,
- genome mutation and structural validity,
- crossover behavior and disabled-gene inheritance,
- species compatibility and complexity calculations,
- neural network activation and bounded outputs,
- simulation sensor vector encoding.

The standard local execution path is `just test`, which invokes CTest with `--output-on-failure`. The same test executable is also run in CI on Linux and Windows.

4.2 Integration Testing

Integration testing is performed by exercising implemented command chains that cross module boundaries. The most important integration flows are:

- config load and validation through `moonai --validate`,
- app startup from CLI arguments through `main.cpp` into `App`,
- headless execution that produces the structured output bundle,
- experiment analysis that converts run artifacts into HTML,
- profiler execution that produces timing JSON and profiler HTML output.

These tests verify that independently correct subsystems work together with the actual file formats and runtime wiring used by the project.

4.3 System Testing

System testing treats MoonAI as a finished executable. The current system scenarios include:

- GUI launch on a machine with a display server,
- headless launch on a machine without visualization,

- CPU-only execution through `--no-gpu`,
- experiment discovery via `--list` and selection via `--experiment`,
- generation of the output directory and final report records,
- publication visibility of generated PDFs through the website document list.

Because the project uses SFML for visualization, the GUI part of system testing is manual. Automated CI covers the build-and-test portion but not UI interaction.

4.4 Performance Testing

Performance testing is implemented through the dedicated profiler executable and helper commands in the project `justfile`. The profiler captures per-frame timing trees, aggregates repeated runs, and produces a self-contained HTML report. The current performance-oriented workflow includes:

- release-build profiler runs through `just profile-run`,
- profiler report generation through `just profile-analyse`,
- full profiler pipeline through `just profile`,
- optional GPU kernel inspection with `just ncu`,
- optional system-level CUDA tracing with `just nsys`.

The collected metrics include average frame time, variability across repeated runs, and timing distribution across major scoped events.

4.5 User Acceptance Testing

Formal external acceptance testing is not implemented in the repository. The current user acceptance process is therefore internal and manual. The participants are:

- Oğuzhan Özkaya,
- Emir Irkılata,
- Caner Aras.

Each participant executes a short set of end-user scenarios on the built application: start the simulation, interact with the visualization controls, run a headless experiment, inspect generated outputs, and confirm that the analysis or profiler reports open correctly.

4.6 Beta Testing

MoonAI does not currently have an external public beta program, public telemetry, or a standalone bug-reporting portal. The implemented beta process is therefore a **closed internal beta smoke pass** performed by the same three team members on a release-candidate build. This internal beta focuses on regression detection, platform sanity, and document publication checks rather than large-scale external feedback.

5 Test Environment

The current MoonAI test environment combines local developer machines and hosted CI executors. The project supports both CPU-only and optional CUDA-enabled execution.

5.1 Software Environment

Table 4: Software environment used by the implemented tests

Category	Tool	Role
Build system	CMake 3.21+	Configures the project, test targets, optional CUDA support, and profiler build.
Build runner	Ninja	Recommended local build backend for fast incremental builds.
Dependency manager	vcpkg	Supplies SFML, GoogleTest, sol2, spdlog, and JSON dependencies used in tests and application builds.
Unit test framework	GoogleTest	Executes C++ unit tests for simulation and evolution logic.
Test launcher	CTest	Standard command-line interface for running discovered tests locally and in CI.
Automation entry point	just	Wraps common build, test, profiling, and analysis commands.
Visualization runtime	SFML 3	Provides the GUI surface used in manual system and acceptance testing.
Analysis runtime	Python + uv	Runs the analysis and profiler HTML report generators.
CI platform	GitHub Actions	Executes cross-platform build and test workflows.
Performance tooling	moonai_profiler, ncu, nsys	Measures frame timing and optional GPU kernel behavior.

5.2 Hardware and Platform Matrix

Current platform coverage is summarized below.

Linux developer machine
: Main local environment for building, running, profiling, report generation, and GUI validation.

Windows GitHub Actions runner
: Automated CI coverage for build and test execution in Debug and Release configurations.

Linux GitHub Actions runner
: Automated CI coverage for build and test execution in Debug and Release configurations.

CUDA-capable Linux machine
: Manual performance and system coverage for the optional GPU path, profiler GPU timing, Nsight Compute, and Nsight Systems workflows.

Headless environment
: Non-visual execution mode used for experiment runs, artifact generation, and report pipelines.

5.3 Test Inputs and Expected Artifacts

The primary runtime input is `config.lua`, optionally combined with CLI flags such as `--experiment`, `--headless`, `--validate`, `--set`, and `--no-gpu`. Successful test execution may generate the following artifacts:

- `config.json` snapshot for the executed run,
- `stats.csv` time-series metrics,
- `species.csv` species-level summaries,
- `genomes.json` representative genome snapshots,
- analysis HTML reports under `analysis/output/`,
- profiler JSON and HTML reports under `output/profiles/` and `profiler/output/`,
- CI logs for automated build-and-test validation,
- the final PDF test-plan report under `docs/TestPlanReport.pdf`.

6 Test Cases

Each test case below is assigned a stable test identifier. The cases are derived directly from implemented repository capabilities and therefore map to existing automated tests, supported commands, or manual smoke procedures.

6.1 Automated Unit Test Cases

TP-UT-001
: **Objective:** Innovation tracking consistency.
Procedure: Execute the same-connection innovation-tracking case from `test_evolution.cpp` through CTest.
Expected result: Repeated requests for the same connection pair return the same innovation identifier within the same mutation window.
Responsible: Team.

TP-UT-002
: **Objective:** Mutation preserves executable genomes.
Procedure: Execute the mutation-validity case from `test_evolution.cpp`.
Expected result: Repeated mutation still produces a genome that can be

converted into a neural network and activated with bounded outputs.
Responsible: Team.

TP-UT-003 **Objective:** Crossover produces valid offspring.
Procedure: Execute the crossover-validity case from `test_evolution.cpp`.
Expected result: The child genome remains structurally valid and produces the expected output count when activated.
Responsible: Team.

TP-UT-004 **Objective:** Sensor vector encoding correctness.
Procedure: Execute the sensor-encoding case from `test_simulation_sensors.cpp`.
Expected result: Predator and prey sensor vectors encode the correct nearest-target deltas, sentinel values, and density metrics.
Responsible: Team.

6.2 Integration and System Test Cases

TP-IT-001 **Objective:** Config validation and override rejection.
Procedure: Run `just validate` and repeat validation with intentionally invalid `--set` values.
Expected result: Valid configurations print `OK`; invalid values report field-specific errors and stop execution.
Responsible: Team.

TP-IT-002 **Objective:** Headless run artifact generation.
Procedure: Run `just run -- --headless --steps 60` and inspect the created output directory.
Expected result: The application completes headlessly and writes `config.json`, `stats.csv`, `species.csv`, and `genomes.json`.
Responsible: Team.

TP-IT-003 **Objective:** Analysis pipeline integration.
Procedure: After a completed headless run, execute `just experiment-analyse`.
Expected result: A self-contained HTML analysis report is generated under `analysis/output/`.
Responsible: Team.

TP-IT-004 **Objective:** Profiler pipeline integration.
Procedure: Build the profiler path and execute `just profile-run`; then execute `just profile-analyse`.
Expected result: Profiler JSON is written under `output/profiles/`

and an HTML report is generated under `profiler/output/`.
Responsible: Team.

TP-ST-001 **Objective:** GUI startup and shutdown.
Procedure: Launch `just run` on a machine with a display server, observe startup, then close the window.
Expected result: The application opens successfully, renders the simulation, and exits cleanly while flushing final output files.
Responsible: Team.

TP-ST-002 **Objective:** CPU fallback path.
Procedure: Execute `just run -- --headless --no-gpu --steps 60`.
Expected result: The simulation runs successfully without CUDA and still produces the standard output bundle.
Responsible: Team.

TP-ST-003 **Objective:** Experiment discovery and selection.
Procedure: Execute `just list-experiments`; then run one named experiment with `--experiment`.
Expected result: The executable lists known experiments and runs the selected one without manual config edits.
Responsible: Team.

TP-ST-004 **Objective:** Cross-platform CI regression.
Procedure: Trigger or inspect the implemented CI workflows for Linux and Windows Debug and Release jobs.
Expected result: Each job configures, builds, and runs the test suite without failure before merge or release packaging.
Responsible: Team.

6.3 Performance, Acceptance, and Beta Test Cases

TP-PT-001 **Objective:** Release-build frame-time profiling.
Procedure: Execute `just profile-run` on a release build and inspect the generated summary report.
Expected result: Average frame time, kept-run counts, and timing charts are produced successfully for the sampled profiler suite.
Responsible: Team.

TP-PT-002 **Objective:** GPU performance inspection.
Procedure: On a CUDA-capable Linux machine, execute `just ncu` or `just nsys`.
Expected result: GPU profiling completes and reports timing information

for the targeted kernel or traced run.
Responsible: Team.

TP-UAT-001 **Objective:** Internal GUI usability smoke pass.
Procedure: Oğuzhan Özkaya, Emir Irkılata, and Caner Aras each run the GUI build, pause or resume, step once, zoom, pan, and select an agent.
Expected result: Core controls respond correctly and the displayed simulation state remains usable for inspection.
Responsible: Oğuzhan Özkaya, Emir Irkılata, Caner Aras.

TP-UAT-002 **Objective:** Internal experiment workflow acceptance.
Procedure: The same three participants execute a named headless experiment, inspect the output files, and open the generated analysis report.
Expected result: A project user can complete the standard experiment workflow without changing the source code.
Responsible: Oğuzhan Özkaya, Emir Irkılata, Caner Aras.

TP-BT-001 **Objective:** Closed internal beta smoke pass.
Procedure: Before final publication, the team repeats the core system, analysis, and document-publication checks on the release-candidate state.
Expected result: No blocking regression remains in build, test, runtime, profiling, or PDF publication paths.
Responsible: Oğuzhan Özkaya, Emir Irkılata, Caner Aras.

7 Control Procedures and Schedule

This section defines how test execution is controlled, when it is considered complete, and how the work is scheduled relative to the report deadline.

7.1 Entry Criteria

Testing may begin when all of the following are true:

- the project configures successfully with CMake,
- required dependencies are available through vcpkg,
- the selected build includes `MOONAI_BUILD_TESTS=ON`,
- the chosen configuration file passes basic syntax loading,
- the intended execution mode is available: display server for GUI tests or headless mode for non-visual tests,
- the team agrees on the test build or release-candidate revision being exercised.

7.2 Exit Criteria

Testing is considered complete for a report submission when the following conditions hold:

- the automated CTest suite completes without failure,
- CI jobs for supported platforms remain green,
- manual system and acceptance smoke cases are completed by the designated team members,
- required runtime artifacts are generated successfully,
- the final PDF report is compiled and placed under `docs/`,
- the website document list exposes the published PDF.

7.3 Failure Handling and Regression Control

MoonAI uses lightweight but concrete control procedures:

- a local or CI test failure blocks the corresponding change from being considered complete,
- fixes are followed by rerunning the failed automated command or repeating the failed manual scenario,
- release packaging is gated by automated tests in the existing release workflow,
- the final report PDF is regenerated after content changes so the published artifact always matches source.

7.4 Test Schedule

The schedule below is defined relative to the Week 9 submission deadline so it remains valid even if the exact calendar date shifts.

Table 5: Relative test schedule

Window	Activity	Responsible
`Deadline minus 14 to 10 days`	Run and stabilize automated unit tests, inspect CI failures, and verify configuration validation paths.	Team
`Deadline minus 10 to 7 days`	Execute headless integration scenarios, artifact checks, and cross-platform runtime sanity checks.	Team
`Deadline minus 7 to 5 days`	Run release-build profiler and optional GPU performance checks on the available CUDA machine.	Team
`Deadline minus 5 to 3 days`	Perform internal user acceptance and closed beta smoke checks with Oğuzhan Özkaya, Emir Irkılata, and Caner Aras.	Oğuzhan Özkaya, Emir Irkılata, Caner Aras
`Deadline minus 3 to 1 days`	Freeze the report, compile the PDF, publish it under `docs/`, update the website entry, and prepare email attachments.	Team
`Deadline day`	Final verification that the PDF opens correctly from the website and matches the submitted version.	Team

8 Roles, Risks, and Deliverables

8.1 Roles and Responsibilities

The project team shares responsibility for implementing, running, and reviewing the current MoonAI test process. Because this is a course project with a small team, responsibilities are intentionally lightweight and collaborative.

Table 6: Testing roles and responsibilities

Responsible personnel	Testing responsibility
Oğuzhan Özkaya, Emir Irkılata, Caner Aras	Maintain and execute the automated unit, integration, system, and performance test commands used by the repository.
Oğuzhan Özkaya, Emir Irkılata, Caner Aras	Perform internal GUI smoke testing, user acceptance scenarios, and the closed internal beta pass.
Oğuzhan Özkaya, Emir Irkılata, Caner Aras	Compile the final PDF report, place it under `docs/`, and verify public website visibility.
Ayşenur Birtürk	Review the report as project supervisor and provide academic guidance on quality and completeness.

8.2 Risks and Mitigations

Table 7: Key testing risks

Risk	Impact	Mitigation
CUDA hardware unavailable	GPU-specific performance and runtime checks cannot be executed on every machine.	Keep CPU fallback test coverage active and run GPU checks only on the available CUDA-capable machine.
GUI requires display access	GUI smoke tests cannot run inside fully headless automation.	Keep GUI checks manual and rely on headless automated paths for CI.
Cross-platform dependency drift	Linux and Windows builds may diverge because of toolchain or dependency changes.	Continue running both platforms in CI for pull requests and release workflows.
Large experiment runtime cost	Full-scale performance or experiment runs may exceed available time before the deadline.	Use shorter smoke runs for test verification and reserve long runs for analysis tasks when time permits.
Small manual tester pool	Internal acceptance and beta coverage are limited to three team members.	Use a repeatable smoke checklist and ensure all three team members run the same scenarios.
Report publication mismatch	The website may expose an outdated PDF if the docs artifact is not refreshed after edits.	Rebuild the PDF immediately before publication and verify the website entry against the generated file.

8.3 Deliverables

The implemented test process produces the following deliverables:

- the LaTeX source of this report under `docs/testplan/`,
- the compiled PDF report under `docs/TestPlanReport.pdf`,
- automated test evidence through local CTest runs and GitHub Actions logs,
- generated runtime artifacts for experiment, analysis, and profiler flows,
- the website document entry that exposes the report publicly.

The required email distribution of the PDF to the supervisor, jury members, and course coordinator is a submission activity external to the repository, so it is tracked as a final project-delivery action rather than a software-generated artifact.

9 Glossary

`CI` Continuous Integration. In MoonAI this refers to the GitHub Actions workflows that configure, build, and test the project automatically on Linux and Windows.

`CTest` CMake's command-line test runner. MoonAI uses it to execute GoogleTest-discovered unit tests locally and in CI.

`CUDA` NVIDIA parallel-computing platform used by MoonAI for optional GPU acceleration.

`Headless mode` Execution mode enabled by `--headless`. The simulation runs without creating the SFML visualization window.

`Internal beta` The closed release-candidate smoke pass executed only by the project team, not by external public users.

`NEAT` NeuroEvolution of Augmenting Topologies, the evolutionary algorithm used by MoonAI for evolving neural network topology and weights.

`Nsight Compute` NVIDIA profiling tool invoked by the `just ncu` recipe for kernel-level GPU performance inspection.

`Nsight Systems` NVIDIA profiling tool invoked by the `just nsys` recipe for CUDA and runtime tracing.

`Profiler suite` A repeated set of timed runs collected by `moonai_profiler` and summarized in an HTML report.

`Smoke test` A short verification run intended to prove that a major workflow still works after recent changes.

UAT User Acceptance Testing. In this project it is implemented as an internal manual check by Oğuzhan Özkaya, Emir Irkılata, and Caner Aras.

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